

# Bhavy Khurana

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## Education

### Manipal Institute of Technology

*B.Tech in Computer Science Engineering (Artificial Intelligence and Machine Learning)*

Manipal, Karnataka

2023 – 2027

**CGPA:** 7.93

**Coursework:** Deep Learning, Operating Systems, Machine Learning, Object-Oriented Programming, Database Management Systems, Big Data Analytics

**Certifications:** Big Data Specialization (UC San Diego)

## Work Experience

### Lepton Software

*Data and Analytics Intern*

2026

- Built a sub-hourly nowcasting pipeline to predict rainfall-driven traffic disruption, combining live persistence with GAN-downscaled 2km precipitation fields for short-horizon road impact forecasting.
- Designed and maintained the geospatial data layer by integrating ERA5 reanalysis, Google Weather API forecasts, and road-network mappings using H3 spatial indexing, PostGIS, and PostgreSQL.
- Backtested the production system on **59,000+** served road-hour predictions, reducing speed-error by **18–63%** versus the climatological baseline and outperforming persistence carry-forward by **33–38%** at 2–3 hour lead times.
- Developed multi-quantile prediction workflows in PyTorch to support uncertainty-aware forecasting and operational decision-making for traffic impact estimation.
- **Tools Used:** Python, PyTorch, spateGAN, Multi Quantile Regression, PostGIS, PostgreSQL

## Projects

### *Hopfield Networks with Adaptive Memory and Dynamic Pattern Forgetting*

- Built a Hopfield-network variant that resists memory degradation under interference by adding feedback-driven adaptive forgetting, proposed as **Adaptive Hopfield Memory (AHM-DPF)**.
- Improved recall accuracy by **25–33 percentage points** over classical Hopfield baselines under high-interference regimes on a purpose-built bit-pattern benchmark (8-D to 64-D).
- **Tools Used:** Python, PyTorch, NumPy, Matplotlib, Pandas

### *Adversarial Robustness of Network Intrusion Detection Systems*

- Built a DNN-based NIDS on UNSW-NB15 achieving a weighted F1 of **0.709**.
- Evaluated adversarial robustness under FGSM & PGD attacks; exposed **81.7%** macro-F1 collapse and a degenerate prediction artifact that inflated robustness estimates in prior work.
- Designed  $\epsilon$ -randomized adversarial training for PGD with balanced loss weighting, restoring adversarial performance to exceed the undefended model's clean-data baseline and eliminating model collapse across realistic perturbation budgets.
- **Tools Used:** Python, PyTorch, Scikit-learn

### *Network Intrusion Anomaly Detection System*

[GitHub](#)

- Built a hybrid intrusion detection pipeline combining a supervised Random Forest layer for known threats with unsupervised One-Class SVM for zero-day attack detection.
- Achieved **0.858 ROC-AUC** for zero-day detection, with the supervised layer evaluated using imbalance-aware metrics on the heavily class-imbalanced UNSW-NB15 dataset.
- **Tools Used:** Python, Scikit-learn, LightGBM, XGBoost, NumPy, Pandas, Matplotlib

## Technical Skills

**Languages:** C, C++, Java, Python, SQL

**Libraries & Tools:** PyTorch, Scikit-learn, Pandas, NumPy, LightGBM, XGBoost, Matplotlib, Git

**Data & Infra:** PostgreSQL, PostGIS, H3 Spatial Indexing